



Smallest 3D transistor

Engineers have developed the smallest 3D transistor yet, by modifying material on an atomic scale. <https://digit.in/may20-81>



World's smallest DRAM

IBM developed the smallest DRAM yet that can provide low power memory for future electronics. <https://digit.in/may20-82>

trolled by passing an electric current through it, and does not need a coil as an intermediary step.

One of the annoying things about smartphones is their tendency to heat up after prolonged use. Although there are checks to prevent any critical failure to the computing elements, overheating can cause the device to malfunction. In some rare and extreme cases, such overheating can cause the batteries to explode. To prevent such situations, engineers put in insulating layers of glass, plastic or even air within the smartphones. All of these occupy space, which is very limited within a handheld device. Researchers from Stanford have developed atomically thin layers of materials that when stacked provide the same insulation as much thicker layers of glass or plastic. The researchers used graphene, molybdenum disulfide and Tungsten diselenide to achieve their results. These heat shields are just two or three nanometers thick, or about 50,000 times thinner than a sheet of paper.

ELECTRONICS BEYOND SILICON

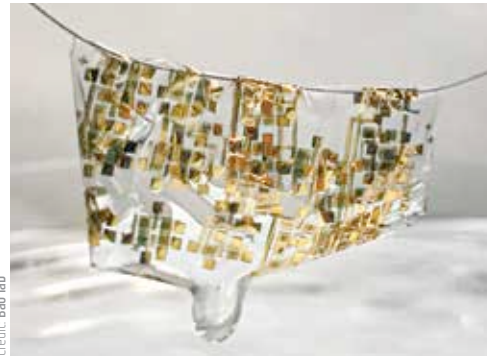
Most of the electronics in the world that we see today, are made up of silicon. Silicon is available in abundance, which is the reason for its widespread use. However, it really is not the optimal material or the best semiconductor available. Silicon transistors are anyway approaching their technological limits, prompting scientists around the world to actively look at alternatives.

There are some materials that are more suitable for electronics, including gallium arsenide, gallium nitride, and lithium fluoride, which are being researched for potential applications in future electronics. Gallium arsenide has been known to researchers for over fifty years, but it was only recently that the material was demonstrated to be compatible with conventional CMOS-based manufacturing processes. Another cheap alternative to silicon is germanium.

So far, the methods of actually creating usable components from

these exotic materials had been expensive. Researchers have developed a method called remote epitaxy, which essentially clones films of the material based on a single wafer, that is expensive to manufacture. Multiple films can be produced from a single wafer. The technique allows for the manufacture of a large number of nanoscale electronics, which have potential applications at scale, such as adding sensors to an entire city.

One of these promising new semiconductors is again, molybdenum disulfide. One of the key barriers to its use was the contact resistance between other metal components and the molybdenum disulfide transistors. Researchers from the University of Purdue worked around



Credit: Bao lab

A flexible, biodegradable semiconductor hung up on a single strand of human hair

this bottleneck by doping the 2D sheets with chemical compounds that reduced the resistance. Molybdenum disulfide is also a lattice structure made up of carbon atoms, similar to graphene, and has many of the same benefits. The advantage it has over graphene is that it is a semiconductor.

STRAIN ENGINEERING

One of the novel techniques that could be implemented in future electronics is called strain engineering. Semiconductors and other crystalline materials have certain properties such as how they conduct heat, conduct electricity, and transmit light. Based on these properties, suitable materials are chosen for use in electronics. The idea is to strain the mate-

rial in a precise manner, that changes the structure of the atoms within, allowing for more desirable properties in the materials. A very slight change can lead to a big advantage. For example, a one percent strain applied in just the right way on silicon chips can increase their electric conductivity by 50 percent. Nanoscale silicon wires can be stretched by 15 percent, and even nanoscale diamond wires can be stretched by 9 percent without affecting their structural integrity. There are other ways to change the properties of materials, such as the chemical doping approach outlined above, but these techniques change the properties of the material permanently. It is also a static change, which cannot be further adjusted. Strain

engineering however, is not permanent. Once the strain is removed, the material returns to its original state. It is also possible to adjust the strain on the materials on the fly, even while it is being used in an electronic device. One of the problems of implementing this approach is just figuring out the properties of the material in all the kinds of strain that can be applied to it. There are three dimensions, along which the strain, that can

be applied outwards, inwards and sideways. Additionally, the strain on the material can be in any angle along three axes. There are just too many possibilities to work out. Researchers from MIT put machine learning to the task, and made computers figure out and predict the amount and direction of strain to apply on a material, to produce a desired target effect. Machines are helping make the next generation of faster machines.

While silicon-based electronics and traditional materials may be approaching their engineering limits, all this research goes to show that the relentless march of technological progress will make sure that Moore's Law continues to be valid for the foreseeable future. 